

Oran Looney

Principal Machine Learning Engineer | Data Scientist

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Professional Summary

Principal Machine Learning Engineer and data scientist with two decades of experience building sophisticated models, applications, and pipelines across insurance and healthcare. Seeking an individual contributor role to directly tackle technical challenges and provide technical leadership.

Experience

QBE — Principal Machine Learning Engineer | 2026–Present

- Spearheaded agentic programming for an in-house document ingestion and generation application for underwriters to reduce dependence on SaaS vendors.
- Developed patterns and processes for building system-of-record applications with coding agents without sacrificing code quality.
- Explored integration of agents into pipelines, structured workflows, and *ad hoc* chat sessions to improve user productivity while adhering to human-in-the-loop best practices.

QBE — Lead Machine Learning Engineer | 2024–2026

- Trained machine learning models to identify high-value insurance submissions based on historical appetite and likelihood to win.
- Led development of a REST API framework serving dozens of insurance pricing models. Introduced logging, semantic versioning, model registration, deployment pipelines, code coverage reporting, performance profiling, and pre-commit hooks to improve quality and uptime.
- Built OCR and LLM pipelines to extract structured information from documents, trained junior data scientists on similar patterns, and designed a benchmark framework to evaluate extraction quality.

Frazier Healthcare Partners — Director, Data Science and AI | 2022–2024

- Developed machine learning classification models for healthcare and life sciences portfolio companies, including models for at-risk student identification, physician survey response probability, and medication adherence analysis.
- Conducted statistical analysis of operational quality metrics and third-party outcomes for ABA therapy, identifying statistically significant relationships between treatment quality, clinician experience, and outcome measures.
- Advised on LLM workflows to identify and flag clinically relevant sections of free-text medical notes, reducing manual medical record review.

Healthgrades — Senior Data Scientist | 2014–2021

- Worked on a large suite of neural network models estimating future healthcare utilization, migrating model training from R to Python and Keras.
- Introduced multi-label training, minority-class oversampling, census data augmentation, hyperparameter optimization, feature evaluation, and inference-time optimization.
- Scaled the training data to ten million records and improved training and inference performance.
- Developed an LDA topic model to group providers into sub-specialties using historical claims data.

- Built a time series model to predict website traffic by geography and category.

Healthgrades — Senior Software Architect | 2013–2014

- Designed and supported software architecture for healthcare data and analytics products.

CPM Marketing Group Inc. — Director of Software Development | 2012–2013

- Led a cross-functional software development team for healthcare marketing analytics applications.

CPM Marketing Group Inc. — Software Engineer | 2007–2012

- Built software systems for healthcare marketing, analytics, and data processing workflows.

Epic Systems — Interface Analyst | 2005–2007

- Supported healthcare system interfaces for enterprise EMR software.

Education

Indiana University — M.S. Physics; M.A. Mathematics | 2004

University of Oregon — B.S. Physics; B.A. Mathematics | 2001

Skills

Machine Learning and Data Science: LLMs, RAG, vector embeddings, neural networks, data drift, generalized linear models, regularized regression, feature engineering, hyperparameter optimization, anomaly detection, topic modeling, time series, random forest, XGBoost, GAM, ARIMA, k-means, PCA, LDA, GMM, MARS

Programming and Data: Python, NumPy, pandas, SciPy, statsmodels, SQL, PostgreSQL, PySpark, Databricks, R, tidyverse, ggplot2, Cython, C, C++, Linux

ML Engineering: PyTorch, Keras, scikit-learn, Hyperopt, FastAPI, MLflow, Docker, AWS, Azure, REST APIs, model serving, model monitoring, deployment pipelines, automated testing, semantic versioning

Leadership and Communication: software architecture, statistical methodology, design of experiments, project management, mentoring, training, scientific communication, machine learning best practices

Selected Writing

Large Language Models: [Vision Models](#), [LLM Benchmarks](#)

Machine Learning: [Induction](#), [Backpropagation](#), [Principal Component Analysis](#)

Statistics: [Logistic Regression](#), [Cross Validated Answers](#)

Numerical Programming: [Genetic Algorithms](#), [Numeric Optimization](#)

Recreational Mathematics: [Angle Addition](#), [Kaprekar Visualization](#), [Bell Numbers](#)